

FAULTY VARIABLE TRANSFORMER

Lessons Learned

This Lessons Learned document focuses on an incident in which a faulty variable transformer (commonly known as a Variac) discharged electricity despite being turned off. This nearly injured a researcher and resulted in equipment damage. It demonstrates the importance of purchasing high-quality equipment from reputable vendors, as well as the importance of verifying electrical systems are de-energized before beginning work.

What Happened?

A lab was preparing to remove fiberglass heating tape from a device. The heating tape was covered by aluminum foil and plugged into an [Ex Electronics Express variable transformer](#) (variable transformers are commonly referred to as Variacs; see the box at right for more details). The Variac was turned off, but was not unplugged.

As the researcher began to remove the foil, there was a large, bright spark of electricity. The researcher was wearing nitrile gloves (to prevent contamination of their equipment, not for protection), and found that their glove had a large scorch mark. After unwrapping the heat tape, the researcher found that it had melted. The researcher unplugged the Variac and removed it from use in the lab.

The lab's PI emailed a local safety contact about the incident. A week later, the safety contact notified EH&S. EH&S requested and received a formal incident report.

Why Did This Happen?

The immediate cause of the incident was a fault in the Variac that caused it to continue to output electricity even when switched off. Reviews on Amazon of this specific model indicated that multiple other purchasers experienced safety issues, and reviewers who opened up the equipment casing found instances of poor soldering, short circuits, and the power switch being connected to the wrong wire.

Additionally, heating tape can become brittle after use, causing wires to break free. These wires can then protrude, causing short circuits if they come in contact with another conductor. The heating tape in this instance had been used previously at ~80% of its temperature limit, which may have caused some embrittlement. Protruding wire then most likely made contact with the aluminum foil, causing a short circuit.

The root cause of this incident was the difficulty in determining equipment safety in online marketplaces. It can be difficult to find reviews warning of safety issues, usually requiring potential buyers to actively search out such information rather than it being readily available. Without clear signs indicating safety issues, purchasers may end up selecting the lowest cost option, unaware of the lower quality. In this case, the lab had not actively searched for information specifically discussing the safety of the product.

Variable Transformers

Variable transformers are devices that provide an adjustable-voltage output of alternating current. Many scientists refer to variable transformers as Variacs, the name of a well-known brand of variable transformers. These days, Variacs are most commonly used in labs to control heating.

What Went Right?

Ultimately, no one was injured and the incident only resulted in minor property damage. Several other factors prevented the incident from being more serious:

- The lab's workspace was free of combustible clutter that could have been ignited by the spark.
- While nitrile gloves are not designed to protect against electricity, in this circumstance the glove provided some protection, but also did not heat up enough to melt (which could have caused an injury).
- The faulty equipment was removed from service.

How to Prevent and Respond to Incidents Like This

EH&S has a number of recommendations for how labs can prevent incidents like this:

- Purchase high-quality equipment for research work.
 - When possible, researchers should purchase NRTL-certified equipment models.
 - When NRTL certification is not available for a class of equipment, researchers should purchase equipment directly from reputable manufacturers when possible, or from reputable vendors (e.g., Grainger, McMaster-Carr).
 - Researchers should read available product reviews, looking for any reports of hazardous malfunctions.
- Unplug equipment before handling wiring rather than just turning it off. Then, verify that the system is de-energized using a multimeter or other voltage probe.
 - If an electrical system contains capacitors, ground them, short them, or wait for them to discharge before handling the system further.
- Inspect electrical equipment carefully before use, looking for exposed wires, cracked or damaged insulation, loose connections, etc.
 - Heat tape in particular should be closely inspected due to its tendency to become brittle.
- Plug Variacs into receptacles equipped with ground fault circuit interrupters (GFCIs). If no GFCI-equipped receptacles are nearby, you can use GFCI adaptors instead.
 - GFCIs cut power if they detect that current has “leaked” from a circuit, including current leakage from someone getting shocked.
- Wear eye protection while working with electrical equipment. While there was no eye injury in this incident, large electrical discharges can cause sparks and debris.
- Report incidents to EH&S as soon as possible by submitting an [incident report](#). You should also call 650-725-9999 at any time for serious incidents (e.g., injury resulting in hospitalization, large hazardous materials spills).